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Abstract

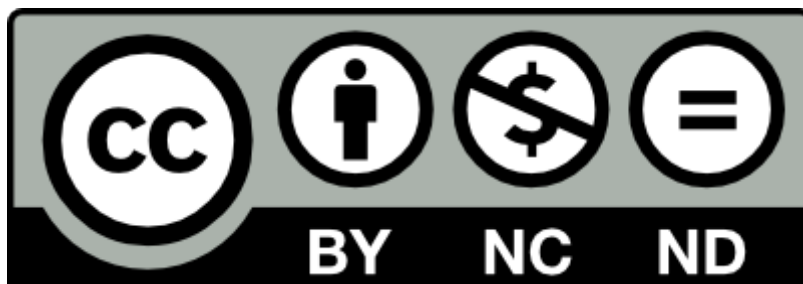
Aim: To analyze the relationship between the depth of the chest compressions and the fluctuation caused in the thoracic impedance (TI) signal in out-of-hospital cardiac arrest (OHCA). The ultimate goal was to evaluate whether it is possible to identify compressions with inadequate depth using information of the TI waveform.

Methods: 60 OHCA episodes were extracted, one per patient, containing both compression depth (CD) and TI signals. Every 5s the mean value of the maxima of the CD, Dmax, and three features characterizing the fluctuations caused by the compressions in the TI waveform (peak-to-peak amplitude, area and curve length) were computed. The linear relationship between Dmax and the TI features was tested using Pearson correlation coefficient (r) and univariate linear regression for the whole population, for each patient independently, and for series of compressions provided by a single rescuer. The power of the three TI features to classify each 5s-epoch as shallow/non-shallow was evaluated in terms of area under the curve, sensitivity and specificity.

Results: The r was 0.34, 0.36 and 0.37 for peak-to-peak amplitude, area and curve length respectively when the whole population was analyzed. Within patients the median r was 0.40, 0.43 and 0.47, respectively. The analysis of the series of compressions yielded a median r of 0.81 between Dmax and the peak-to-peak amplitude, but it decreased to 0.47 when all the series were considered jointly. The classifier based on the TI features showed 90.0%/37.1% and 86.2%/43.5% sensitivity/specificity values, and an area under the curve of 0.75 and 0.71 for the training and test set respectively.

Conclusion: Low linearity between CD and TI was noted in OHCA episodes involving multiple rescuers. Our findings suggest that TI is unreliable as a predictor of Dmax and inaccurate in detecting shallow compressions.

Keywords: Automated external defibrillator (AED); Cardiac arrest; Cardiopulmonary resuscitation (CPR); Chest compression (CC); Compression depth (CD); Thoracic impedance (TI).



Can thoracic impedance monitor the depth of chest compressions during out-of-hospital cardiopulmonary resuscitation?

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Methods: 60 OHCA episodes were extracted, one per patient, containing both compression depth (CD) and TI signals. Every 5 s the mean value of the maxima of the CD, $D_{\max}$, and three features characterizing the fluctuations caused by the compressions in the TI waveform (peak-to-peak amplitude, area and curve length) were computed. The linear relationship between $D_{\max}$ and the TI features was tested using Pearson correlation coefficient (r) and univariate linear regression for the whole population, for each patient independently, and for series of compressions provided by a single rescuer. The power of the three TI features to classify each 5 s-epoch as shallow/non-shallow was evaluated in terms of area under the curve, sensitivity and specificity.

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Keywords

Cardiac Arrest, Thoracic Impedance (TI), Compression Depth (CD), Chest Compression (CC), Automated External Defibrillator (AED), Cardiopulmonary Resuscitation (CPR), CPR Quality

1. INTRODUCTION

High quality chest compressions (CCs), i.e. compressions given with an adequate rate, depth, and full chest recoil, are crucial for the restoration of spontaneous circulation during cardiopulmonary resuscitation (CPR)¹. Several studies have shown that increasing compression force or depth during CPR increases blood flow in animals²⁻⁴ and end-tidal CO₂ levels in humans⁵. Moreover, in out-of-hospital cardiac arrest (OHCA), increasing compression depth (CD) is associated with higher chance of survival to hospital discharge⁶⁻⁸. Current resuscitation guidelines recommend applying CCs with the force necessary to depress the sternum at least 50 mm⁹.

Shallow CCs are common both in hospital and out-of-hospital cardiac arrest^{6,7,10,11}. Rescuer fatigue causes a decrease in depth over time, although the CC-rate remains constant. To overcome this problem, several mechanisms based on additional sensors have been developed for real-time CPR feedback, some of which have been integrated into automated external defibrillators (AEDs). Although they have been shown to be effective⁶, they require accessory devices.

The thoracic impedance (TI), acquired through the defibrillation pads of an AED, has been suggested as an alternative method that could be used to provide CPR feedback. This signal is recorded by injecting an alternating current between electrodes and measuring the resulting voltage. Fluctuations induced in the TI waveform by CCs have been successfully used to mark the instants of CCs^{12,13}, to measure CC-rate^{13,14}, and to detect CC interruptions^{15,16}. Moreover, manual/automatic review of episodes for quality evaluation in terms of CC-rate and CC-fraction is possible based on the TI¹². However, the relationship between the TI fluctuations and the CD has not been established yet.

A recent study by Zhang et al.¹⁷ reported strong correlation between the TI amplitude change and both the CD and the coronary perfusion pressure in a porcine model of cardiac arrest. In a controlled scenario with 2 rescuers and 14 similar animals, they concluded that TI could serve as a surrogate marker of CD. Contradictory results have been reported for humans in short communications. Brody et al.¹⁸ reported greater TI fluctuations for stronger CCs, while Aramendi et al.¹⁴ showed low correlation between CD and TI amplitude fluctuations. However, limited details of the analytical methods as well as database used were published in these two studies. The main purpose of this study was to analyze the relationship between features of TI fluctuations and CD

31 in episodes recorded during OHCA, where multiple rescuers were involved in the resuscitation
32 effort. Linear relationship was first evaluated for the whole population, then, within patients, and
33 finally, for series where one patient and a single rescuer were involved. The discriminative power
34 of the features of the TI was also evaluated. The final question to be answered is whether there is
35 information in the TI waveform that permits identifying CCs with inadequate depth in OHCA.

2. MATERIALS AND METHODS

2.1. Data materials

The dataset used in this study was a subset of a large cardiac arrest registry containing 623 OHCA episodes acquired by the Tualatin Valley Fire & Rescue service (Tigard, Oregon, USA). The episodes, one per patient, were collected using the Philips HeartStart MRx monitor/defibrillator between 2006 and 2009. From the original dataset only 189 episodes had both the CD and the TI signals, and we extracted the 60 having both signals uninterruptedly recorded along the whole episode and a minimum of 1000 CCs. The CD signal (sampling rate 250 Hz) was computed from the force and the acceleration recorded through a CPR assist pad. The TI signal was recorded through the defibrillation pads by applying a sinusoidal excitation current (32 kHz, 3 mA peak-to-peak) with a resolution of 0.74 mΩ per least significant bit, a bandwidth of 0-80 Hz and a sampling rate of 200 Hz. For the analysis, isolated CCs that were not part of a group with at least 10 CCs were discarded, so a total of 116679 CCs were analyzed.

Every episode of the dataset reflected the resuscitative efforts of multiple rescuers. Crews composed of 2-6 rescuers performed 2-min CC series with stops in-between to analyze the rhythm and rotate between them. To analyze intervals where the single-rescuer-single-patient pattern was guaranteed, we defined CC series as epochs of CCs performed without interruptions by a single rescuer on a patient. Interruptions in CCs longer than 1.5 s were identified as a possible change of rescuer¹⁹. Only series with a minimum duration of 2 min were considered for the analysis.

2.2. Signal processing and feature extraction

The CD and TI signals were processed to compute the maximum depth of each CC and the features of the fluctuation caused by the CC in the TI waveform. First, the CD signal was processed applying a negative peak detector with a static threshold of -15 mm to automatically detect the instant of maximum depth in each CC. The depth of the CC in that instant is denoted as $D_{\max i}$. Taking into account that instant, the cycle of each CC was identified in the CD signal, as shown in Fig. 1, where every CC cycle is depicted as a gray band. Then, the TI signal was downsampled to 100 Hz and band-pass filtered (order 6 Chebyshev filter with 0.1 dB of peak-to-peak ripple in the 0.6-7 Hz passband) in order to remove baseline drift, fluctuations caused by ventilations, and high frequency noise. The processed TI signal during the i^{th} CC cycle was denoted as $z_{pi}[n]$, where n is the time sample number.

To characterize the fluctuations induced by CCs in the TI signal, three different waveform features were computed for each CC:

- **Peak-to-peak amplitude, Z_{ppi} :** Difference between the maximum and the minimum value of $z_{pi}[n]$.

- **Area, A_i :** Area of $z_{pi}[n]$:

$$A_i = \sum_{n=1}^N |z_{pi}[n]| \quad (1)$$

where N is the length in samples of $z_{pi}[n]$.

- **Curve length, C_i :** Length of the curve of $z_{pi}[n]$, computed as follows:

$$C_i = \sum_{n=2}^N \sqrt{\left(\frac{1}{f_s}\right)^2 + (z_{pi}[n] - z_{pi}[n-1])^2} \quad (2)$$

where f_s is the sampling frequency of the TI signal.

Fig. 1 illustrates two examples where the extracted features are depicted on the CD signal (top) and on the processed TI signal (bottom). Panel a shows an example where the TI fluctuations caused by CCs are quite sinusoidal, while in panel b the fluctuations are more irregular. The area and the curve length were computed because the peak-to-peak amplitude was not able to characterize irregular TI fluctuations (see Fig. 1).

In order to smooth the values of the features, every 5 s the mean values for the CD maxima, $D_{\max}$, and for the three TI features (Z_{pp} , A , and C) were computed.

2.3. Analysis of the linear relationship

The linear relationship between $D_{\max}$ and the TI features was tested using Pearson correlation coefficient (r) and univariate linear regression for the whole population, for each patient independently, and for each series of compressions provided by a single rescuer. The performance of Spearman's rank correlation coefficient was very similar to r in our analyses, but r was used as it permitted comparison with previous studies.

The linear relationship between $D_{\max}$ and Z_{pp} in a single-rescuer-single-patient pattern was analyzed with two different sets of series. First, in order to guarantee a wide range of $D_{\max}$, series

with a minimum standard deviation of 7 mm in the $D_{\max}$ were considered. A single series per patient was selected, the one with the largest standard deviation. A total of 9 series were analyzed. Univariate linear regression was used to predict $D_{\max}$ using Z_{pp} , and r computed independently for each series and jointly for the complete set of series. Second, aiming to replicate the procedure that Zhang et al.¹⁷ used with animals, series with optimal and suboptimal CCs were selected. A series was considered suboptimal when at least 75% of the CCs were below 38 mm, and optimal when at least 75% of the CCs were above 50 mm. One series per patient was selected. When more than one series of the same patient met the criteria, the one with the largest number of CCs above (optimal group) or below (suboptimal group) the threshold was selected. A total of 12 series were jointly analyzed by computing r and fitting a model to the data through univariate linear regression.

2.4. Discrimination power of the TI features

The power of the three TI features to classify each 5 s-epoch as shallow/non-shallow was evaluated in the context of the 2005 resuscitation guidelines²⁰, considering non-shallow CCs those with $D_{\max} \geq 43$ mm and shallow CCs those with $D_{\max} \leq 38$ mm. These limits were proposed for a possible feedback system incorporated into an AED which would alarm the rescuer if the compressions were too shallow. A gap of 5 mm between the limits was introduced to take into consideration measurement errors. Probability density functions for the three TI features were estimated for both classes. After testing for normality, median values for both classes were compared using the Wilcoxon rank sum test.

A multivariate logistic regression model was applied for the classification based on a three feature vector. The regression model assigns to each 5 s-epoch a probability of non-shallow depth given by:

$$P_{\text{non-shallow}}(z) = \frac{1}{1 + e^{-z}} \quad (3)$$

with $z = \theta_0 + \theta_1 \cdot Z_{\text{pp}} + \theta_2 \cdot A + \theta_3 \cdot C$

The 60 episodes were split into training (40) and test (20) sets. The regression coefficients (θ_i) were calculated for the training set using the Waikato Environment for Knowledge Analysis (WEKA) software. The discriminative power of the classifier was evaluated in terms of AUC (Area Under the Curve), sensitivity and specificity for the training and test sets. Sensitivity was defined as the capacity to correctly identify shallow CCs and specificity as the capacity to correctly identify

112 non-shallow CCs. The weight of the classes was set to maximize the specificity while ensuring a
113 sensitivity above 90% in order to provide confident feedback with shallow CCs. $P_{\text{non-shallow}} > 0.5$
114 were classified as non-shallow.

3. RESULTS

Results are presented as median (Q_1 - Q_3), where Q_1 and Q_3 are the 1st and 3rd quartiles respectively, as their distributions did not pass the one-sample Kolmogorov-Smirnov normality test.

The 60 episodes lasted 2068 (1490 – 2473) s and included 1852 (1362 – 2556) CCs per episode. From those episodes a total of 75 series were extracted with a duration of 149 (133 – 187) s; every series had 277 (243 – 347) CCs with a depth of 42 (37 – 48) mm.

Fig. 2 shows the scatter-plot of $D_{\max}$ against each of the TI features for the whole population composed by 14424 values with a depth of 42 (37 – 48) mm. A high dispersion was observed for the three features around the model fitted using univariate linear regression. The value of r was 0.34, 0.36, and 0.37 for Z_{pp} , A , and C , respectively.

The analysis within patients yielded a r of 0.40 (0.24 – 0.66) for Z_{pp} , 0.43 (0.26 – 0.66) for A , and 0.47 (0.25 – 0.68) for C . Fig. 3 shows the scatter-plot and the model fitted for the Z_{pp} in six representative patients. The cases depicted in panels a and b show a high r (0.92 for both cases), although the slopes of their regression lines differ from one patient to another. Panels c and d show two patients with a low r and a high dispersion around the regression line. Finally, in the patients depicted in panels e and f two patterns can be observed, and a single model does not adequately adjust to all the values.

Fig. 4 shows the scatter-plot for the 9 series with a minimum standard deviation of 7 mm in the $D_{\max}$. Values for each series, 28 (26 – 40) per series, are depicted in a different color as well as the fitted model. The black line represents the model fitted to all the series when considered jointly (320 values with a $D_{\max}$ of 48 (40 – 56) mm). Individually, the series presented a r of 0.81 (0.51 – 0.83), but it decreased to 0.47 when all the series were considered jointly.

In Fig. 5 the scatter-plot for the set of 12 optimal and suboptimal series is presented; a total of 390 values, with a $D_{\max}$ of 57 (54 – 63) mm and a Z_{pp} of 3.0 (2.5 – 3.7) Ω for the optimal group, and a $D_{\max}$ of 32 (30 – 34) mm and a Z_{pp} of 0.9 (0.6 – 1.5) Ω for the suboptimal group. The model fitted to the data is represented as a black line, and r was 0.87.

The discrimination analysis was performed with a non-shallow group, $D_{\max}$ of 48 (45–53) mm, composed of 4426 and 1909 values corresponding to 5 s-epochs in the training and test sets respectively, and with a shallow group, $D_{\max}$ of 34 (31–36) mm, with 2773 and 1174 values in the

145 training and test sets respectively. Wilcoxon rank sum test reported significant differences between
146 medians for the three TI features for both groups (p-value < 0.01). The values were 1.5 (1.0 – 2.3)
147 vs 1.1 (0.7 – 1.5) for Z_{pp} , 50 (33 – 77) vs 34 (23 – 49) for A and 3.4 (2.2 – 5.0) vs 2.4 (1.7 – 3.2)
148 for C , for non-shallow and shallow groups respectively. Fig. 6 shows the probability distribution
149 functions for shallow (red) and non-shallow (blue) CCs. Although there were significant differences
150 between medians, a high overlap between distributions was observed.

151 The logistic regression classifier showed 90.0%/37.1% sensitivity/specificity values, and AUC =
152 0.75 for the training set. For the test set sensitivity, specificity and AUC were 86.2%, 43.5% and
153 0.71, respectively.

4. DISCUSSION

Our study is a systematic and retrospective review of complete OHCA episodes representative of the population of patients and rescuers involved during CPR. The linear relationship between the CD and the features of the TI waveform fluctuation was analyzed, and the power of the TI to discriminate shallow CCs quantified.

In short communications, Di Maio et al.²¹ and Brody et al.¹⁸ showed that the TI reflected greater amplitude fluctuation for deeper/stronger CCs with porcine/human models respectively. Nevertheless, in none of the experiments did they have objective measurements of the CD. In 2012 Di Maio et al.²² included information of the CD obtained from a CPR assist pad and their preliminary results for a subset of CCs showed a promising correlation ($r = 0.75$) between CD and TI fluctuations. Unfortunately, this short communication did not provide details on the analytical methods or the dataset used, so their results are not easily comparable to this study.

Our study included a set of episodes with a wide variety of patients and rescuers. A total of 60 patients, with presumably random physical characteristics, and between 2-6 rescuers attending each OHCA event were analyzed. We used three features to characterize the fluctuation caused by the CCs in the TI waveform (Z_{pp} , A and C). The 14424 values obtained for each feature showed very low correlation with D_{max} ($r < 0.38$ in any case). The clouds of the three scatter-plots in Fig. 2 demonstrate that predicting D_{max} with any of the features is highly unreliable. For instance, for a Z_{pp} of 1Ω the probability of error in the predicted value of D_{max} is high because of the wide range of corresponding D_{max} values.

The large dispersion observed in Fig. 2 was next studied by analyzing the episodes independently in order to suppress the variability introduced by different patients. The median r per episode was around 0.43 for any of the three parameters, slightly above that obtained for the whole population. However, the variability between patients was large with a r of 0.40 (0.24–0.66). Fig. 3 shows the scatter-plot of 6 distinct OHCA events where a different slope of the regression line is observed for every patient, which may reflect the individual patient-electrode characteristics. Sex, chest size and body mass have been reported to affect the baseline TI^{23–26}, and TI amplitude fluctuations caused by ventilations have been shown to be correlated with the thoracic fat and the thoracic circumference²⁷. Fig. 3 also shows different dispersion with respect to the regression line from one patient to another. On the one hand, with little dispersion and high slope in the

184 regression line the r is high as can be observed in panels a and b. On the other hand, with high
185 dispersion and low slope the correlation coefficient is low (panels c and d). Panels e and f, however,
186 depict patients where two tendencies can be clearly distinguished. This could be linked to two
187 distinct patterns for providing CPR by the rescuers involved in each patient. In both cases, it is
188 clear that a single regression model would hardly fit all of the values.

189 The impact of several rescuers was eliminated in the series analysis shown in Fig. 4, where
190 the single-patient-single-rescuer pattern was maintained in every series. With a single rescuer the
191 dispersion within each series clearly decreased, resulting in an increased linearity between $D_{\max}$
192 and Z_{pp} (median $r = 0.81$). Nevertheless, the variety of the physical characteristics of the 9
193 patients provided a low correlation, $r = 0.47$, when all the series considered jointly, emphasizing
194 the inability of defining a confident linear fit.

195 In a recent study, Zhang et al. investigated the relationship between TI changes and both
196 CD and coronary perfusion pressure in a porcine model of cardiac arrest¹⁷. In 14 pigs suboptimal
197 (35 mm) and optimal (50 mm) CCs were provided by 2 experts during 2 minutes. The peak-through
198 amplitude change of the TI during each CC was averaged every 5 s, obtaining a feature similar
199 to our Z_{pp} . They found high correlation ($r = 0.89$) between their feature and $D_{\max}$, and great
200 differences in TI amplitude between optimal and suboptimal depth groups (1.45 ± 0.37 vs 0.47 ± 0.12 ,
201 $p < 0.001$) with $D_{\max}$ (44.16 ± 4.61 vs. 29.06 ± 4.90 mm $p < 0.001$).

202 Trying to replicate this study in humans, a set of 12 series with optimal and suboptimal CCs
203 was extracted from different patients. The difference of Z_{pp} for both groups was significant (3.0
204 ($2.5 - 3.7$) Ω vs 0.9 ($0.6 - 1.5$) Ω , $p < 0.001$) with $D_{\max}$ (57 ($54 - 63$) mm vs. 32 ($30 - 34$) mm
205 $p < 0.001$) for optimal and suboptimal depth groups respectively. The scatter-plot obtained in Fig.
206 5 for $D_{\max}$ and Z_{pp} is very similar to the corresponding figure in the animal experiment. Although
207 in both cases the correlation coefficient is high (0.89 for animals and 0.87 for humans), for a given
208 value of Z_{pp} high variability of the predicted $D_{\max}$ was evident. For a proper interpretation of the
209 large correlation coefficient observed, we should take into account the limitations of these analyses.
210 The selection of optimal and suboptimal classes introduces a selection bias, and reduces sample
211 size. When the complete range of compression depths is considered, the correlation coefficient
212 drops to 0.34 as can be observed in Fig. 2. The set of patients and rescuers included was small
213 (12 patients/12 rescuers and 14 animals/2 rescuers). When a greater variety was included, 60

214 patients/2-6 rescuers each, a high dispersion in the scatter-plot was observed (Fig. 2), the r
215 decreased substantially and a given value of Z_{pp} corresponded to a wide range of D_{max} .

216 In this study the TI has proved unreliable to predict the D_{max} , but from a practical perspective
217 the power to discriminate shallow CCs was also analyzed. There is evidence that CCs with a
218 depth of 40 mm or below reduces the percentage of patients admitted alive to hospital^{6,7}. The
219 2005 resuscitation guidelines recommended CCs with depth in the range of 38 – 50 mm²⁰. In our
220 study 69% of the CCs were above the recommended minimum threshold. We tried to discriminate
221 the group of CCs with $D_{max} < 38$ mm from the group with $D_{max} > 43$ mm. The three TI features
222 showed distributions with different medians ($p < 0.001$) but high overlap between the two classes.
223 The logistic regression classifier revealed an AUC = 0.71 with the test set, which implied 86.2%
224 sensitivity for detecting shallow CCs but a 43.5% specificity. It was therefore impossible to safely
225 identify shallow CCs using TI.

226 5. Conclusions

227 The linear relationship between CD and TI in OHCA episodes involving a wide variety of
228 patients and multiple rescuers is poor. With our data TI cannot reliably predict D_{max} and is
229 inaccurate in discriminating shallow CCs.

230 Conflict of interest

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Figure Legends

- Figure 1 Features extracted for two segments of the CD signal (upper panels) and the corresponding $z_p[n]$ signal (lower panels). For each CC cycle, represented as a gray band, the maximum depth, D_{maxi} , is represented on the CD, and the three TI features on $z_p[n]$.
- Figure 2 Scatter-plots of the D_{max} with respect to the TI features Z_{pp} , A and C (from top to bottom respectively) for the whole population. The fitted regression line is depicted in black.
- Figure 3 Scatter-plot and linear model adjusted to the feature Z_{pp} in six patients. Panels a and b represent cases where the model fits the data well and the correlation coefficient is high. Poor fitting and poor correlation is observed for cases in panels c and d due to the high dispersion of the data. Finally, in panels e and f two different patterns may be identified.
- Figure 4 Scatter-plot for the 9 series with a minimum standard deviation of 7 mm in the D_{max} . The values and the regression line for each series are represented in a different color. The regression line fitted to all the data is depicted in black.
- Figure 5 Scatter-plot and regression line of D_{max} vs. Z_{pp} for the set of 12 optimal and suboptimal series. The regression line is depicted in black.
- Figure 6 Probability density functions for shallow (red) and non-shallow (blue) groups for Z_{pp} , A and C from top to bottom respectively.

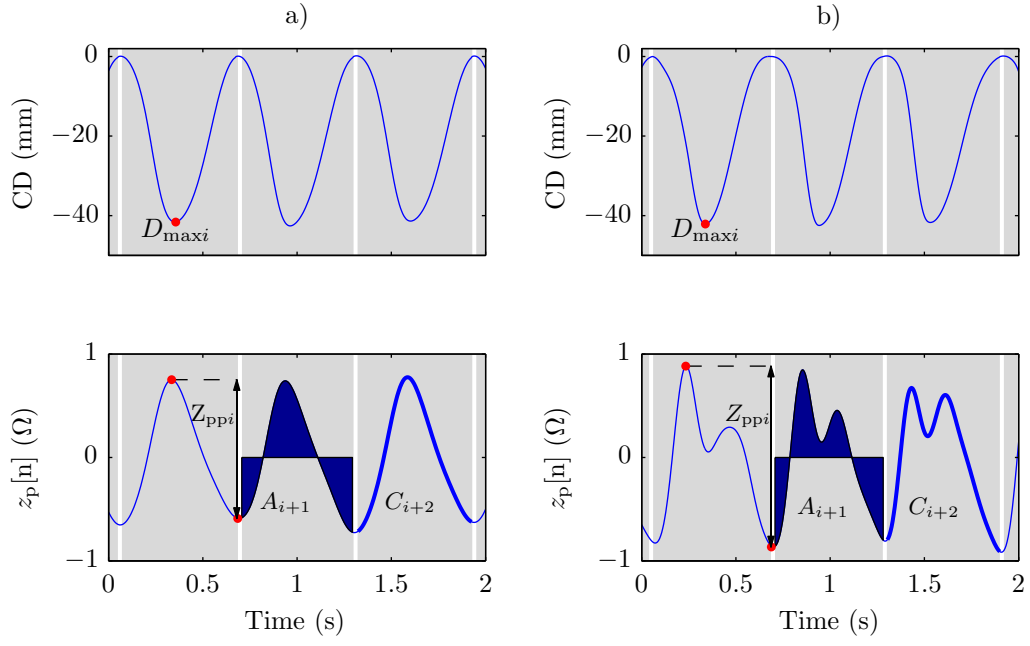


Figure 1: Features extracted for two segments of the CD signal (upper panels) and the corresponding $z_p[n]$ signal (lower panels). For each CC cycle, represented as a gray band, the maximum depth, $D_{\max i}$, is represented on the CD, and the three TI features on $z_p[n]$.

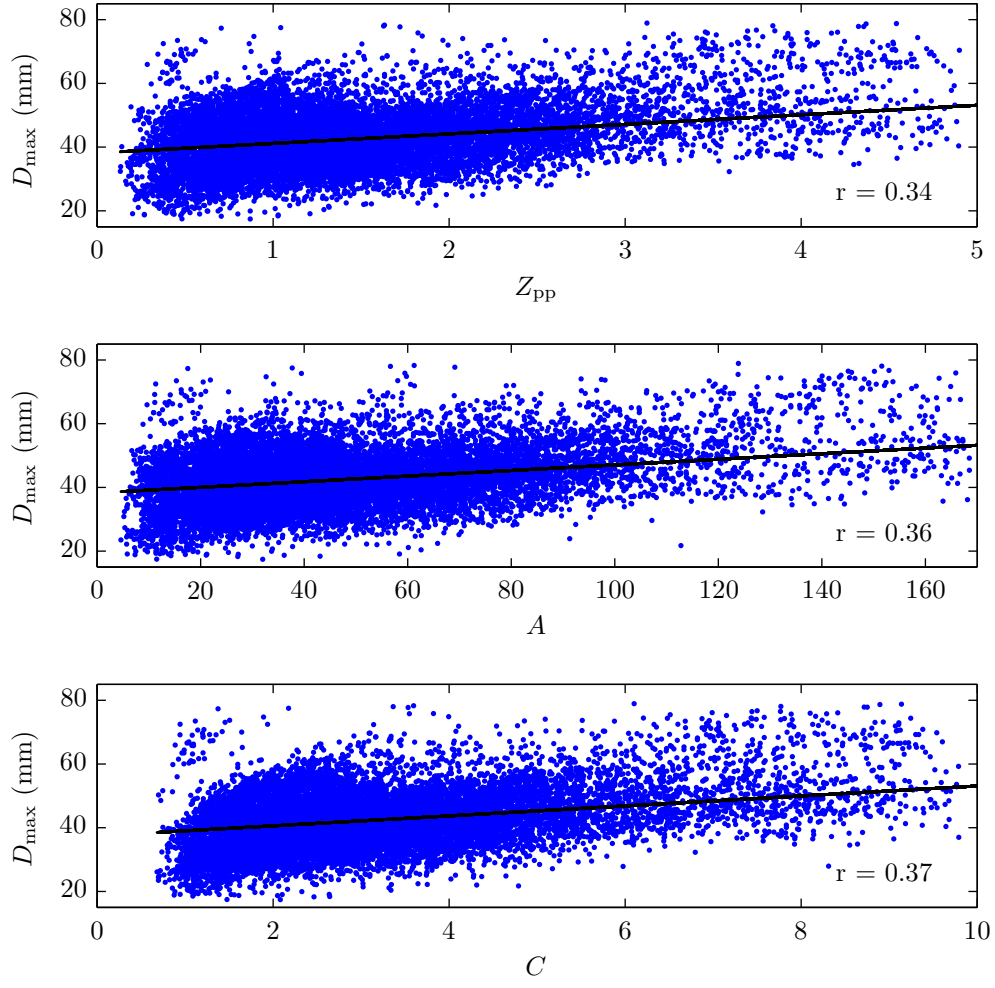


Figure 2: Scatter-plots of the $D_{\max}$ with respect to the TI features Z_{pp} , A and C (from top to bottom respectively) for the whole population. The fitted regression line is depicted in black.

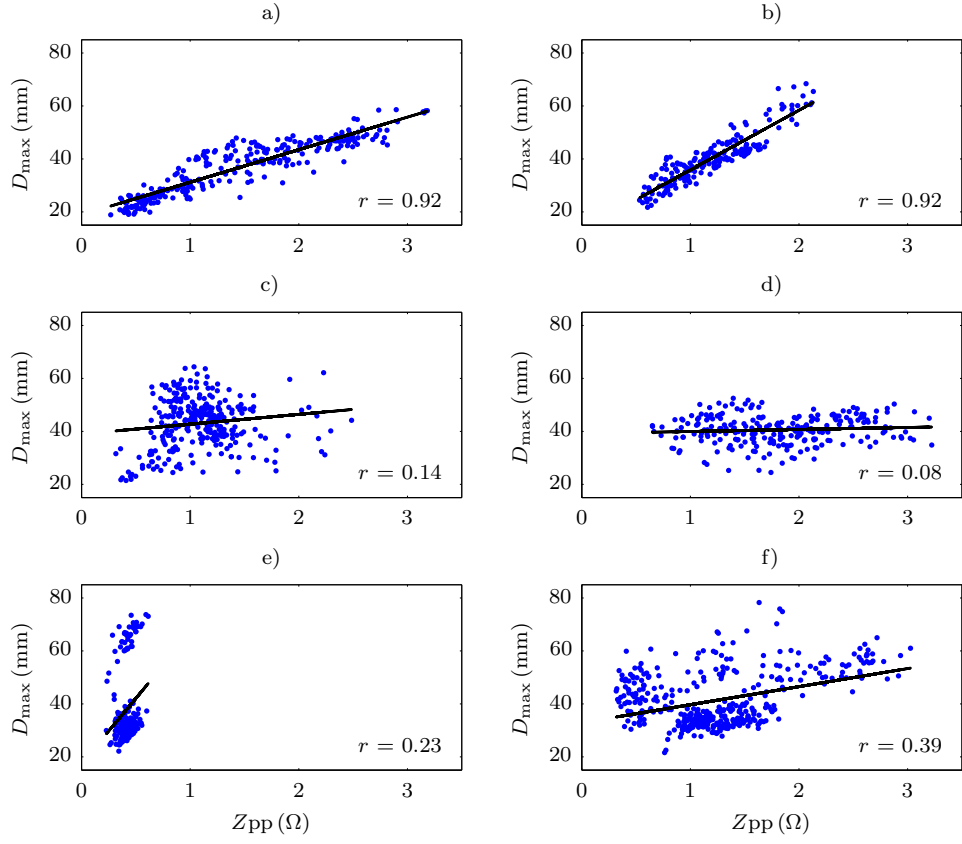


Figure 3: Scatter-plot and linear model adjusted to the feature Z_{pp} in six patients. Panels a and b represent cases where the model fits the data well and the correlation coefficient is high. Poor fitting and poor correlation is observed for cases in panels c and d due to the high dispersion of the data. Finally, in panels e and f two different patterns may be identified.

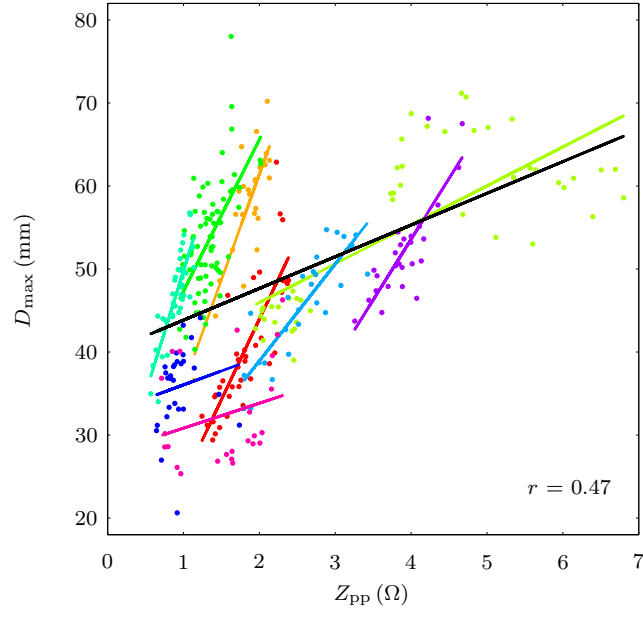


Figure 4: Scatter-plot for the 9 series with a minimum standard deviation of 7 mm in the $D_{\max}$. The values and the regression line for each series are represented in a different color. The regression line fitted to all the data is depicted in black.

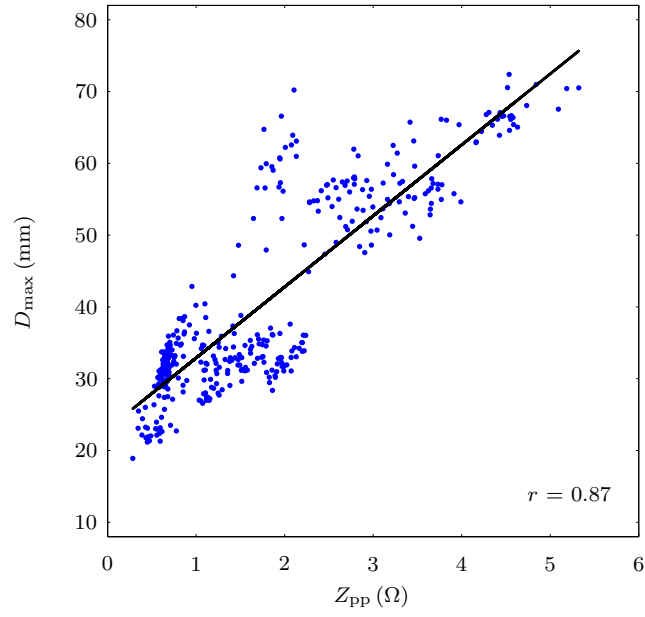


Figure 5: Scatter-plot and regression line of $D_{\max}$ vs. Z_{pp} for the set of 12 optimal and suboptimal series. The regression line is depicted in black.

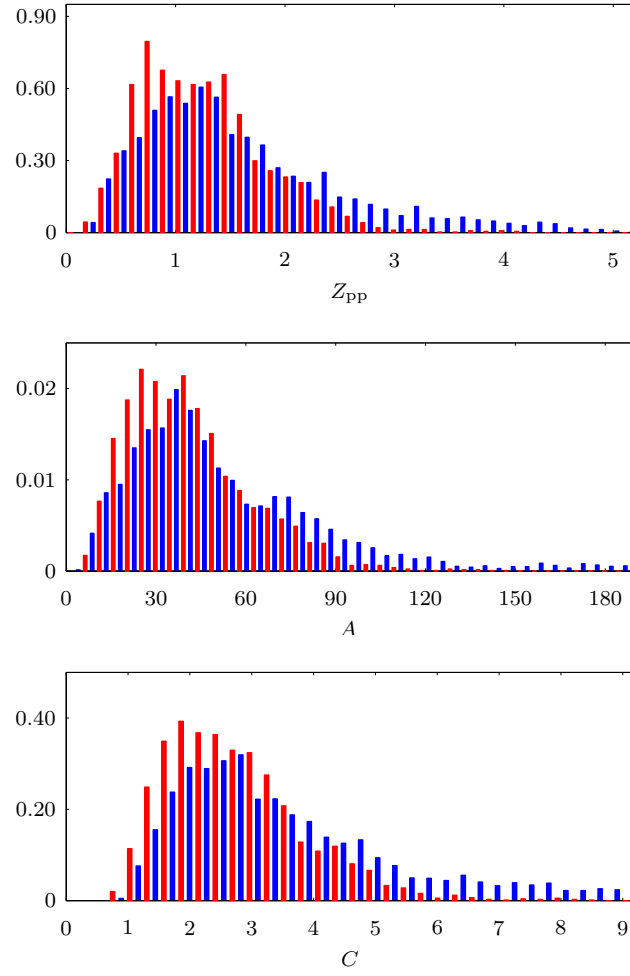


Figure 6: Probability density functions for shallow (red) and non-shallow (blue) groups for Z_{pp} , A and C from top to bottom respectively.